NIST FRAMEWORK

NEED

The NIST Cybersecurity Framework is needed to provide a structured, risk-based approach for managing cybersecurity threats by helping organizations identify, protect, detect, respond, and recover from cyber incidents.

TIME



ROADMAP

- **Diagnose** the following:
  - NIST Framework
  - The National ICT Development Agenda
  - The National ICT Ecosystem Framework.
  - Vision of NICTEF
  - Framework Elements
  - NICTEF as the national Blueprint
  - Aligning the NIST CSF with the national agenda
  - Applicability in the Signal Corps

Experiment:

Reflect:

Evaluation: Through written examination (Essay)

LEARNING OUTCOME

At the end of this subject, the students will be able to discuss the NIST Framework in an operational environment IAW PAM 6-0102-4. (C3)

REFERENCE

- PAM 6-00 (Command and Control Communications and Cyber Systems)
- PAM 6-0102 – Signal Operations
- PAM 3-00 (Army Operations)
- PAM 6-0102-4 Cyberspace Operations
- FM 3-12- Cyberspace and Electronic Warfare Operations

NIST Framework

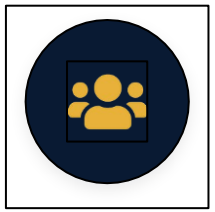
A voluntary, risk-based **playbook** — not a checklist.

Published by the U.S. National Institute of Standards and Technology (NIST), the Cybersecurity Framework (CSF) is a common language for managing and reducing cybersecurity risk. It describes outcomes to achieve, not specific products to buy — so any organization can apply it.

- **Risk-based —** Prioritise effort by likelihood and impact, not by fear.
- **Technology-neutral —** Works across any sector, mission, or size of unit.
- **Interoperable —** Maps to ISO 27001, COBIT, and other control catalogues.
- **A shared language —** Lets commanders, engineers, and partners speak as one.

NIST Framework

Scope of the Framework



WHO IT'S FOR

- Any organization — regardless of sector, size, or maturity
- Government agencies, the AFP, and defense enterprises
- Critical-infrastructure operators (power, telecom, water)
- Private firms, academe, and small units alike



WHAT IT COVERS

- Cybersecurity risk as an enterprise-wide concern
- People, processes, and technology together
- The full lifecycle: govern, protect, detect, recover
- Supply-chain and third-party (C-SCRM) risk



WHAT IT IS NOT

- Not a mandatory law or a one-size compliance audit
- Not a product, tool, or specific technical control
- Not a replacement for your own risk judgment
- Not exhaustive — it complements existing standards

Why the Framework matters

Common
language across leaders, engineers & allied partners

Risk-first
focus that puts scarce resources where they count

Resilience
built in — the mission continues under attack

Trust
with citizens, coalition partners & industry

Turns cyber into a command decision	Frames risk in terms leadership understands, enabling deliberate, defensible choices on where to invest.
Enables interoperability and shared defense	A shared model lets units, agencies, and coalition partners plan, exercise, and respond together.
Strengthens readiness and resilience	Balancing all six Functions means you can absorb a hit, keep operating, and recover fast.

SIX FUNCTIONS of NIST CSF



GOVERN · IDENTIFY · PROTECT

GOVERN

Establish and monitor the cyber risk strategy

- Organizational context & risk appetite
- Roles, responsibilities & authorities Policy and continuous oversight
- Cyber supply-chain risk management (C-SCRM)

IDENTIFY

Understand what you must defend

- Asset management — hardware, software, data
- Risk assessment of threats & vulnerabilities
- Improvement from lessons and reviews. Prioritisation aligned to the mission

PROTECT

Put safeguards in place

- Identity, authentication & access control
- Awareness and role-based training
- Data & platform security, encryption
- Resilient technology infrastructure

DETECT · RESPOND · RECOVER

DETECT

Find trouble quickly

- Continuous monitoring of networks & assets
- Adverse-event analysis & correlation
- Establish baselines to spot anomalies Feed alerts to the response team

RESPOND

Act on confirmed incidents

- Incident management & triage
- Analysis to scope and understand the event
- Reporting and command communication Containment and mitigation actions

RECOVER

Restore and improve

- Recovery-plan execution and restoration
- Verified return to normal operations Communication with stakeholders
- Capture lessons to harden defenses

The three components of the NIST CSF



THE CORE

What to do

A hierarchy of Functions → Categories → Subcategories describing the cybersecurity outcomes to pursue. The 'what,' independent of any specific technology.



PROFILES

Where you stand

A Current Profile captures today's posture; a Target Profile captures the desired state. The gap between them becomes your prioritised action plan.



TIERS

How rigorously

Four Implementation Tiers gauge how disciplined and adaptive your risk-management practice is — from Partial (1) to Adaptive (4).



In practice: use the Core to pick outcomes, Profiles to measure the gap, and Tiers to judge how mature your practice needs to be for the mission. Together they turn cyber risk into a concrete, trackable plan.

The four Implementation Tiers

1

PARTIAL

Ad hoc, reactive risk management; cyber handled case by case.

2

RISK INFORMED

Risk practices approved by management but not yet org-wide policy.

3

REPEATABLE

Formal policies, consistently applied and updated as threats change.

4

ADAPTIVE

Continuous improvement; lessons and threat intel reshape defenses.

Increasing rigour, agility & integrati on →

Source of Army Standards

Applying the Framework in seven steps

1. **Prioritize & scope** Define the mission, systems, and units in scope.
2. **Orient** Identify assets, threats, and regulatory drivers.
3. **Create a Current Profile** Assess which outcomes you achieve today.
4. **Conduct a risk assessment** Judge likelihood and impact of key risks.
5. **Create a Target Profile** Define the desired outcomes for the mission.
6. **Analyze gaps & prioritize** Compare profiles; build a prioritized action plan.
7. **Implement & repeat** Execute, monitor, and refine continuously.

IN A MILITARY CONTEXT

- Treat the Target Profile as commander's intent for cyber.
- Map Functions to mission-essential tasks and systems.
- Use Tiers to set readiness levels per operational need.
- Align profiles across units for interoperable defense.
- Fold RESPOND/RECOVER into continuity-of-operations plans.

INTRODUCTION TO NICTEF

THE NATIONAL ICT DEVELOPMENT AGENDA

- From 1992 to 1998, the “National Information Technology Plan for the 21st Century” was made mainstream by the Department of Transportation and Communications (DOTC), the Department of Trade and Industry (DTI), and the National Computer Center (NCC). The plan envisioned the Philippines to be “a newly industrialized country” and “Asia’s Knowledge Centre

INTRODUCTION TO NICTEF

THE NATIONAL ICT ECOSYSTEM FRAMEWORK (NICTEF)

- The government's linchpin role in embracing innovation builds upon ICT as the currency of the digital economy. And for this objective, the NICTEF will be an authoritative reference source, as well as a living document that shall be updated periodically with the latest plans, programs and projects, including recent indicators, accomplishments, outcomes and results

What is the NICTEF?

National ICT Ecosystem Framework

Issued by the Department of Information and Communications Technology (DICT), the NICTEF is the country's overarching roadmap for ICT development.

It succeeds the earlier Philippine Digital Strategy and is anchored to the Philippine Development Plan and AmBisyon Natin 2040.

ITS TWIN PURPOSE

- **Promote the national ICT development agenda; and**
- **Serve as a blueprint for the harmonization and coordination of national ICT plans, programs, and projects.**

WHY IT MATTERS FOR DEFENSE

One national direction
Aligns fragmented ICT efforts across agencies into a coherent whole.

Cyber as a pillar
Trust and security of ICT are treated as foundational, not optional.

Shared infrastructure
Reliable connectivity that military networks also depend upon.

Whole-of-nation
A basis for civil-military coordination on ICT and cyber.

The NICTEF vision

VISION *A digitally empowered, innovative, globally competitive and prosperous society where everyone — especially the poor and vulnerable — has reliable and affordable access to and use of ICT.*

Vibrant ICT ecosystem

Reliable, affordable, and inclusive connectivity nationwide.

Digital government

Efficient, citizen-centered public services delivered online.

Trust & security in ICT

Cybersecurity and data privacy that build public confidence.

Innovation & competitiveness

A digital economy that grows jobs and global standing.

INTRODUCTION TO NICTEF

FRAMEWORK ELEMENTS

- ICT is a game-changer, transforming the way people do things. Today, cities are dynamic societies powered by innovation and driven by knowledge. With this, the Department recognizes the challenges and opportunities that make up the framework elements

INTRODUCTION TO NICTEF

FRAMEWORK ELEMENTS

The National ICT Ecosystem is comprised of several interdependent framework elements:

- **Human Capital:** Talents and Skills – the individuals who access the applications, services, content and data that are provided by the players in the ICT ecosystem
- **Affordable Access and Devices:** The interfaces through which humans access the applications, services, content and data – these may include wearable devices, cellphones, laptops, desktop computers, internet cafes, and other similar devices or venues

INTRODUCTION TO NICTEF

FRAMEWORK ELEMENTS

The National ICT Ecosystem is comprised of several interdependent framework elements:

- **Platforms (Apps/Services and Content/Data)** – these are the solutions (or portions of a solution) which are accessed by users in the ecosystem in order to achieve equitable, inclusive, and sustainable development in our society, and potentially to improve their quality of life
- **Infostructure/Infrastructure** – these are the physical and logical components which collectively perform the function of providing secure connectivity between the users, their devices, and the platform which they are accessing

INTRODUCTION TO NICTEF

FRAMEWORK ELEMENTS

The National ICT Ecosystem is comprised of several interdependent framework elements:

- **Standards, Regulations and Policies** – these provide the boundaries which will allow for the players and elements within the ecosystem to safely and productively inter-connect and inter-operate

INTRODUCTION TO NICTEF

THE FIVE FRAMEWORK ELEMENTS



NICTEF as the national blueprint

The framework's core job: pull scattered ICT plans, programs, and projects into one coordinated national effort.

Agency ICT plans

Infrastructure projects

e-Gov programs

Cybersecurity plans

NICTEF
harmonize & coordinate

No duplication
Overlapping projects and spend are avoided.

Interoperability
Systems and data connect across agencies.

Measured progress
Shared targets make outcomes trackable.

Stronger cyber posture
Security is coordinated, not siloed.

Aligning the NIST CSF with the national agenda

NIST CSF 2.0

The HOW

- Operational method for managing cyber
- risk Six Functions, Profiles, and Tiers
- Unit- and system-level implementation

NICTEF

The WHAT & WHY

- National ICT direction &
- priorities Harmonizes plans
- across agencies

Sets the strategic, whole-of-nation context

The synthesis: NICTEF sets the national direction and the National Cybersecurity Plan sets policy; the NIST CSF gives units a proven, repeatable method to actually deliver those outcomes at the tactical and technical level.

Applicability to the Signal Corps

Secure the network backbone
Apply PROTECT and DETECT to tactical comms, transmission systems, and command networks the Signal Corps operates.

Run continuous monitoring
Stand up DETECT capabilities — sensors, baselines, and analysis — as a core cyber-battalion function.

Drill response C recovery
Rehearse RESPOND and RECOVER as part of COOP so operations survive and bounce back from intrusion.

Govern with command intent
Use GOVERN to set clear cyber roles, policy, and oversight aligned to higher headquarters.

Interoperate & align
Share Target Profiles across units and align to NICTEF and the National Cybersecurity Plan.

Build the human edge
Sustain awareness and technical training — the decisive factor in every Function.

CHECK ON LEARNING

SUMMARY

- ✓ Introduction to NIST and NICTEF
- ✓ Strategic Trends in the ICT Ecosystem



THANK YOU!



HAVE A NICE DAY

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BEAT BOUTIQUE